PROJECT REFLOW: DIGITAL SUPPLY CHAIN TRANSFORMATION

Executive Boardroom Strategy Briefing | NorthStar Retail Group ($1.2B Enterprise)

A deterministic engineering roadmap resolving intra-facility bottlenecks and transport fragmentation.

CAD $13.0M

CAPITAL EXPENDITURE

Leaves $7.0M Cash Reserve

CAD $8.08M

ANNUAL NET VALUE

Recurring Pre-Tax Savings

+$7.094M

3-YEAR NPV (10% WACC)

Superior Capital Return

3.5%

LATE DELIVERY TARGET

Down from 8.5% Baseline

THE CAD $1.2B SUPPLY CHAIN CHALLENGE

Three regional DCs fulfilling 12.0M customer orders annually are experiencing structural margin drag.

High shipping volumes mask severe underlying cost inefficiencies that threaten NorthStar's omnichannel margins.

12.0M Orders

ANNUAL VOLUME

40,000 Orders / Day Across 3 DCs

$86.4M OpEx

LOGISTICS COST

$7.20 Cost Per Delivered Order

8.5% Late

DELIVERY FRICTION

1,020,000 Late Shipments Annually

18,000 Tonnes

CARBON FOOTPRINT

Scope 1 Diesel Fleet Emissions

ISOLATING ROOT CAUSES FROM SURFACE SYMPTOMS

A rigorous MECE issue tree demonstrates that 90% of logistics friction occurs in picking and trunking.

Surface symptoms (customer complaints) stem directly from intra-DC search drag and dead-head transport miles.

38% Travel Drag

WAREHOUSE PICKING

Manual searching drives 2.4% error rate

62% Fill Factor

TRANSPORT FLEET

LTL fragmentation wastes trailer space

12-Mo Build Trap

MICRO-HUB EXPANSION

Option C adds $2.2M recurring OpEx

Zero Debt Risk

CAPITAL STRATEGY

Preserve $7.0M cash contingency buffer

DETAILED EVALUATION OF THREE TRANSFORMATION LEVERS

Analyzing autonomous warehouse robotics, algorithmic transportation ML, and regional micro-hubs.

Option A and Option B offer immediate operational synergy; Option C introduces capital-heavy execution drag.

Option A ($8.0M)

WAREHOUSE COBOTS

$4.9M net savings | -2.0% late | 19.6 mo

Option B ($5.0M)

TRANSPORT ML

$3.9M net savings | -3.2% late | 15.4 mo

Option C ($12.0M)

REGIONAL HUBS

$3.8M net savings | 37.9 mo payback

Capital Ceiling

$20.0M BUDGET

Board mandates strict 3-year payback

EXHAUSTIVE PERMUTATION EVALUATION & RANKING

Ranking every feasible combination of Options A, B, and C against capital, service, and ESG mandates.

Portfolio A+B emerges as the undisputed champion, delivering maximum NPV while preserving $7.0M in liquidity.

A+B: CHAMPION

CAD $13.0M CAPEX

+$7.094M NPV | 19.3 mo payback

Option B: Ranked #2

CAD $5.0M CAPEX

+$4.698M NPV | High ROI but small

B+C: Sub-Optimal

CAD $17.0M CAPEX

+$1.528M NPV | Crushed by hub OpEx

Option C: Rejected

CAD $12.0M CAPEX

37.9 mo payback breaches 3-year rule

PORTFOLIO A+B: ROBOTIC PICKING & TRANSPORT ML

A dual-lever operational strategy combining facility automation with national dynamic freight clustering.

Synchronized execution ensures fast cash generation in Month 4, fully funding downstream facility robotics.

Pillar 1: Cobots

DC 1 & 2 ROBOTICS

Vision-guided picking slashes cycle time 38%

Pillar 2: Routing ML

FLEET OPTIMIZATION

Trailer cube fill expands from 62% to 84%

Pillar 3: Cash Buffer

CAD $7.0M RESERVED

35% of authorized budget kept as cash

Timeline

12 MONTHS TOTAL

Full $8.08M run-rate achieved by Month 6

DETERMINISTIC 3-YEAR CASH FLOW & DCF SCHEDULE

Baby-step financial proofs showing zero skipped arithmetic and rigorous 10.0% WACC discounting.

Portfolio A+B generates $8.08M in annual recurring pre-tax savings, paying back the full $13.0M capex in 19.3 months.

Year 0 Outflow

(CAD $13,000,000)

Total upfront capital expenditure

Year 1 Pre-Tax

CAD $8,080,000

Discounted PV: CAD $7,345,455

Year 2 Pre-Tax

CAD $8,080,000

Discounted PV: CAD $6,677,686

Year 3 Pre-Tax

CAD $8,080,000

Discounted PV: CAD $6,070,624

THREE-SCENARIO STRESS TESTING (BEAR / BASE / BULL)

Proving capital safety under adverse operational and macroeconomic market conditions.

Even in the Bear Case (-20% savings, +10% capex), Portfolio A+B generates +CAD $1.76M in 3-year NPV.

Bear Case (-20%)

CAD $14.3M CAPEX

+$1.76M NPV | 26.6 mo payback

Base Case (Plan)

CAD $13.0M CAPEX

+$7.09M NPV | 19.3 mo payback

Bull Case (+25%)

CAD $13.0M CAPEX

+$12.11M NPV | 15.4 mo payback

Downside Safety

$7.0M BUFFER

No debt needed under any scenario

DECARBONIZATION & QUANTITATIVE THROUGHPUT GAINS

Translating digital algorithms and cobot assistance into physical supply chain acceleration.

4,500 tonnes of annual CO2e eliminated (-25.0%), meeting corporate ESG commitments ahead of schedule.

Late Rate Cut

8.5% -> 3.5%

600,000 fewer late deliveries annually

Trailer Utilization

62% -> 84%

940,000 road miles eliminated annually

Carbon Reduction

-4,500 TONNES

Scope 1 diesel emissions cut by 25.0%

Picker Productivity

+62% UPTIME

Walking distance cut from 12 km to 1.5 km

PHASED EXECUTION ACROSS THREE DISCIPLINED HORIZONS

Horizon 1 deploys software routing for immediate cash flow; Horizon 2 and 3 scale facility robotics.

Governance dashboard establishes dual-track predictive leading indicators and financial lagging metrics.

Horizon 1 (M1-M4)

TRANSPORT ML

$5.0M Capex | Late rate -3.2 pts | Fuel -18%

Horizon 2 (M3-M6)

DC 1 & 2 COBOTS

$5.5M Capex | Full $8.08M run-rate at M6

Horizon 3 (M7-M12)

DC 3 AUTOMATION

$2.5M Capex | Enterprise load balancing

Accountability

VP LOGISTICS & OPS

Monthly board governance milestones

2x2 RISK MATRIX & PROACTIVE MITIGATION CONTROLS

Pre-mortem stress testing addressing software integration, equipment delivery, and carrier adoption.

Retained CAD $7.00M contingency cash buffer insulates program against unexpected cost spikes.

Cobot Integration

LOW PROB / HIGH IMP

Off-peak pilot staging in DC 1

Hardware Slippage

MED PROB / MED IMP

Software-first H1 buffers hardware delay

Carrier Buy-In

LOW PROB / MED IMP

Fuel rebate incentives & API integration

Budget Overrun

LOW PROB / LOW IMP

$7.0M cash reserve covers +53% capex shock

UNANIMOUS STRATEGIC RECOMMENDATION & NEXT STEPS

Immediate executive motion to authorize CAD $13.0M for Portfolio A+B execution across 3 Horizons.

Project ReFlow achieves superior capital returns, fixes core operational friction, and elevates brand equity.

Capital Authorization

CAD $13.0M

Releases Horizon 1 tranche ($5.0M)

Liquidity Reserve

CAD $7.0M

Preserved as uncommitted board buffer

Year 1 Cash Flow

+$8.08M / YR

First algorithmic cash inflows at Month 4

Board Decision

UNANIMOUS APPROVAL

Executive sponsorship assigned